A PROJECT REPORT ON

SMART RESTAURANT USING AI

BY

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Project Report submitted to Department of Computer Science, Government College Kasaragod in partial fulfillment of the requirements for the award of Degree of BSc Computer Science.

GOVERNMENT COLLEGE KASARAGOD
(Affiliated to KANNUR UNIVERSITY)
Vidyanagar , Kasaragod – 671123
(2021-2024)

DECLARATION

We hereby declare that the project work titled “SMART RESTAURANT USING AI” written and submitted by us is our original work. We also declare that this report has not been submitted to any other Universities or Institutions for the award of any fellowship, degree or diploma.

By,

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CERTIFICATE

This is to certify that the project “SMART RESTAURANT USING AI” submitted in partial fulfillment of the requirement of the degree of BSc Computer Science is a result of bonafide work carried out “ AMRUTHA BALAN ”, during the academic year 2023-2024.

Internal Project Guide

External Examiner

Head of the Department

1.

2.

ACKNOWLEDGEMENT

We are happy to submit our project "SMART RESTAURANT USING AI" where we try to improve scenario in restaurants and hotels for better customer experience. We take this opportunity to express our sincere gratitude to the people who have helped us in our attempt. We wish to express our heartfelt thanks to our principal Dr. V S Anil Kumar, our Head of department Ms. Akhila P, Assistant Professor, Department of Computer science and our project guide Ms. Akhila P Assistant Professor Department of Computer Science. We are extremely thankful to all the staffs of the Computer Science department for their kind co-operation and encouragement to make the project success. Finally, we would like to thank the team faculty and our lab instructors who extended their helping hand on various occasions.

Sincerely,

AMRUTHA BALAN

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ABSTRACT

Innovative AI technologies are transforming the restaurant industry, offering solutions to traditional system limitations. Existing systems often lack personalized recommendations hindering customer satisfaction and operational effectiveness. Our proposed AI-enabled Smart Restaurant utilizes dynamic recommendations generated through Convolutional Autoencoder analysis. This innovative system enhances customer satisfaction, ensuring optimal dining experiences and driving business growth in the competitive hospitality sector. Leveraging a Convolutional Autoencoder for food recommendation provides advantages in capturing intricate patterns within food-related data. Its spatial relationship preservation, non-linear feature learning, and dimensionality reduction capabilities make it suitable for extracting meaningful representations, enhancing the precision of the recommendation system.

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1.INTRODUCTION

1.1 OVERVIEW

In many popular restaurants, waiters/waitresses tend to miss out on tables or customers' calls during busy hours potentially decreasing ones clientele. While this is an ongoing issue, there is still no product that drastically improves the communication between the servers and the customers in the current market. Hence, the goal is to design a system in which the customers can call their servers easily and help the restaurant increase overall efficiency. An internal wired communication system will allow prompt notification to the server when a customer requires service. Moreover, servers can also be more focused on serving their current customers and save their time and energy from always keeping an eye out for needy customers. On top of meeting the needs of customers, restaurant managers can also monitor the response time of their waiters/waitresses through us of this system. Hardworking, proficient employees will become more recognized while lazy, inefficient employees become motivated to improve. As a result, the restaurant becomes more efficient and possibly increasing morale while improving the level of customer satisfaction.

1.2 OBJECTIVE

Now a day, the world is getting into digital world. System was trying to build out to make people even more convenience on any aspects. Smart restaurant provides an system based food ordering system without help of a waiter. It involves seller and buyer which could bring benefits to each other. It is win-win situation that will bring benefits to the world. This system was bringing a convenience for customer that can make order by help of a computer.

It provides customer with a completely new way to make order. By providing customer convenience and also increase the sales. No time wasted with order taking or letting the customer browse the menu over the system.

This system provides more reliable usability, maintainability and dependability functions. By creating quality, easy to keep track management of new system. So that it can streamline all the works by a simple click. By develop the system based food ordering system can reduce the unnecessary costs such as staff salaries, customer satisfaction, reputation, etc. Our restaurant also easy to keep track for the maintenance and point redemption.

1.3 PROBLEM STATEMENT

Design a recommendation system using Convolutional Auto Encoder.

1.4 PROJECT DESCRIPTION

Here, in this smart restaurant we had built our own dataset named “smart dataset” it contains 5100 rows. Our dataset appears to consist of four columns, with one column representing items and the remaining three columns containing different food combinations where a combination1, combination2, combination3 corresponding to each item. It likely captures the relationships between items and various food combinations associated with them. Analyzing this dataset could provide insights into preferred or common pairings of foods with specific items.

1. Data Preprocessing:
    - Clean and preprocess the transactional dataset to extract meaningful information about user-item interactions
    - Construct an adjacency matrix based on co-occurrence or user-item interaction frequencies.
2. Model Architecture:
    - Design a convolutional autoencoder tailored for processing adjacency matrices.
    - The encoder part of the autoencoder will transform the high-dimensional adjacency matrix into a lower-dimensional representation.
    - The decoder will reconstruct the original adjacency matrix from the learned representation.
3. Training:
    - Train the convolutional autoencoder using the adjacency matrix dataset.
    - Optimize the model parameters to minimize the reconstruction loss between the original and reconstructed adjacency matrices.
4. Evaluation:
    - Assess the quality of the learned representations by evaluating the reconstruction accuracy of the autoencoder.
    - Validate the effectiveness of the learned representations in capturing meaningful product features by visualizing embeddings or using clustering techniques.
    - Evaluate the recommendation performance using metrics such as precision, recall.
5. Recommendation Generation:
    - Utilize the trained encoder portion of the autoencoder to generate low-dimensional representations of products.

- Implement a recommendation algorithm that compares the similarity between product representations to generate top-k recommendations for users.

1.5 MOTIVATION

In the evolving landscape of the hospitality industry, the traditional restaurant model faces challenges that hinder customer satisfaction and operational efficiency. The motivation behind this project is rooted in the desire to revolutionize the dining experience by leveraging cutting-edge technologies. By introducing a smart restaurant system with QR code scanning for menu access, AI-driven recommendations for personalized dining, and an advanced payment method, the project aims to address the limitations of conventional restaurant systems. The goal is to provide customers with a seamless and engaging dining experience, meeting the growing expectations for contactless interactions, personalized service, and efficient payment methods. Additionally, the project seeks to empower restaurant owners with tools to streamline their operations and enhance overall business performance. Through this innovative approach, the project aims to contribute to the transformation of traditional dining establishments into modern, tech-savvy hubs that cater to the evolving preferences and expectations of both customers and restaurant owners.

2. EXISTING SYSTEM

Today when we go to a restaurant, we need to search all the items in the menu for the similar items upon a selected item. But in our smart restaurant, when we select a item, we get a recommend regarding it.

Having a smart restaurant that provides recommendations based on selected items definitely enhances the dining experience by making it more convenient and personalized. It saves time and helps customers discover new dishes they might enjoy.

3. PROPOSED SYSTEM

Using AI we recommend three food combinations according to selected item. This is done by using a convolutional auto encoder. The convolutional autoencoder for recommendation utilizes deep learning techniques to generate personalized suggestions. By encoding input data into a low-dimensional representation, it captures intricate patterns and relationships. In this context, the model takes adjacency matrices representing product interactions or ingredient co-occurrences. The encoder compresses this high-dimensional data into a latent space, while the decoder reconstructs the original matrix. Through training, the autoencoder learns to preserve relevant features for recommendation tasks. The resulting representations enable effective recommendation generation by comparing similarities in the latent space. This approach facilitates tailored suggestions, enhancing user experience in various domains, from e-commerce to culinary exploration.

3.1.2 METHODOLOGY

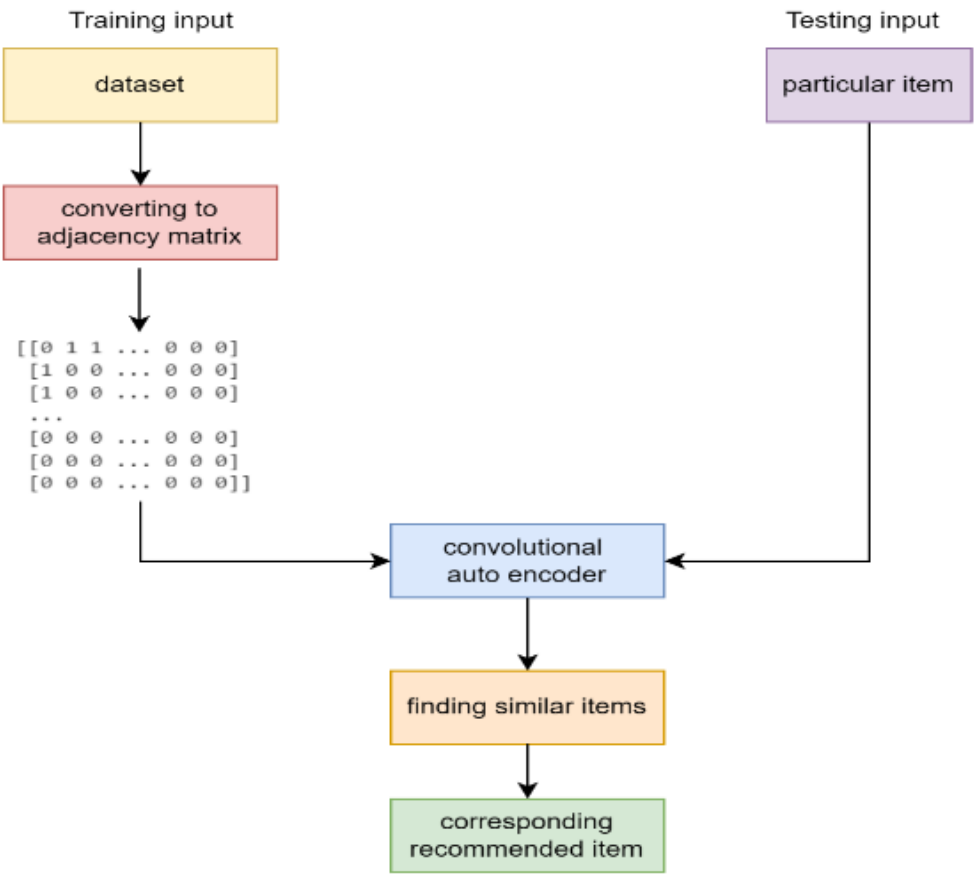


Fig 3.1 Methodology diagram

Dataset:

The training input dataset typically consists of user-item interactions or preferences. Each entry in the dataset would represent a user interacting with or expressing a preference for an item. When a user selects one item, the system recommends another item that is somehow related or correlated to the selected item.

Converting to adjacency matrix:

The adjacency matrix is a way to represent these relationships in a mathematical form. Each row and column in the matrix corresponds to an

item, and the entries in the matrix represent the strength of the relationship between items.

Particular item:

Testing input typically consists of interactions between users and items that were not used during the training phase. This could include data such as user-item interactions that occurred after the training period or a subset of interactions that were held out from the training set for evaluation purposes. Here particular item is given as testing input.

Convolutional autoencoder:

Using a convolutional autoencoder for recommendation systems is an innovative approach that leverages deep learning techniques to capture complex patterns in user-item interactions.

Finding similar items:

It involves comparing attributes or features of items to identify those with shared characteristics or patterns. This process can be crucial in recommendation systems for suggesting relevant alternatives to users based on their interactions.

Recommended items:

After training and testing, recommendations are generated by inferring user preferences through the trained model, ranking items based on predicted scores, and presenting the top-N recommendations to users. Continuous monitoring and improvement ensure high-quality recommendations, adapting to evolving user preferences over time.

3.1 SYSTEM ARCHITECTURE

As mentioned earlier we are using a CONVOLUTIONAL AUTO ENCODER model to implement AI Recommendation. After training, we get the k-recommendation according to the selected item .

3.1.1 Convolutional Autoencoder

In this recommendation system architecture, the process begins with the extraction of product relationships from a dataset, typically indicating which products are frequently purchased together or share some other form of association. These relationships are encoded into an adjacency matrix, where each row and column represent a specific product, and the matrix elements denote the strength of the relationship between pairs of products. This adjacency matrix serves as the input to a Convolutional Autoencoder (CAE), a specialized variant of autoencoder tailored for graph-structured data. The CAE consists of two main components: an encoder and a decoder. The encoder component compresses the high-dimensional adjacency matrix into a lower-dimensional feature space while preserving essential relational information. This process is facilitated by convolutional operations that capture local patterns in the graph structure. The decoder, conversely, reconstructs the original adjacency matrix from the learned low-dimensional representation, aiming to minimize the reconstruction error between the input and output matrices during training. Through this iterative process, the CAE learns to encode meaningful representations of product relationships into a lower-dimensional feature space, facilitating efficient storage and computation while retaining crucial information for recommendation tasks.

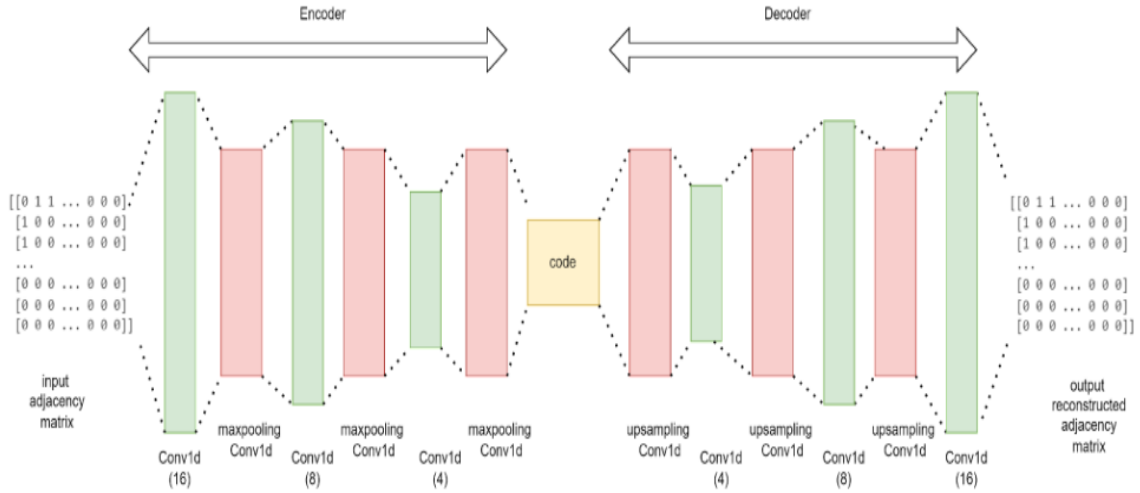


Fig 3.2 Architecture of convolutional autoencoder

```
(4896, 4896)
Model: "model"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 input_1 (InputLayer)        [(None, 4896, 4896, 1)]   0

 conv2d (Conv2D)             (None, 4896, 4896, 32)    128

 max_pooling2d (MaxPooling2  (None, 2448, 2448, 32)    0
 D)

 conv2d_1 (Conv2D)           (None, 2448, 2448, 64)    6208

 conv2d_2 (Conv2D)           (None, 2448, 2448, 128)   24704

 max_pooling2d_1 (MaxPoolin  (None, 1224, 1224, 128)   0
 g2D)

 conv2d_3 (Conv2D)           (None, 1224, 1224, 64)    24640

 up_sampling2d (UpSampling2  (None, 2448, 2448, 64)    0
 D)

 conv2d_4 (Conv2D)           (None, 2448, 2448, 32)    6176

 up_sampling2d_1 (UpSamplin  (None, 4896, 4896, 32)    0
 g2D)

 conv2d_5 (Conv2D)           (None, 4896, 4896, 1)     97

=================================================================
Total params: 61953 (242.00 KB)
Trainable params: 61953 (242.00 KB)
Non-trainable params: 0 (0.00 Byte)
```

Fig 3.3 Summary of the model

During the testing phase, the trained CAE is deployed for recommendation purposes. Specifically, only the encoder component is utilized, as it effectively distills the product relationships into a compact feature vector representation. When presented with a target product, its corresponding feature vector is obtained by passing the product's row in the adjacency matrix through the encoder. This feature vector encapsulates the product's relationships with other items in the dataset condensed into a lower-dimensional space. To generate recommendations for the target product, similarity measures such as cosine similarity or Euclidean distance are computed between the target product's feature vector and those of all other products. The products with the highest similarity scores are then selected as the top recommendations for the target product. Typically, the system would output a ranked list of recommended products, with the top-ranking items deemed most relevant based on their similarity to the target product's learned representation. By leveraging the power of convolutional autoencoders and graph-based representations, this approach offers a principled and efficient framework for generating personalized recommendations in various e-commerce and recommendation system scenarios.

```
[[0 1 1 ... 0 0 0]
 [1 0 0 ... 0 0 0]
 [1 0 0 ... 0 0 0]
 ...
 [0 0 0 ... 0 0 0]
 [0 0 0 ... 0 0 0]
 [0 0 0 ... 0 0 0]]
```

4893x4893

Fig 3.4 Adjacency matrix

The number of unique dishes across all rows determines the size of the matrix. In this matrix there are 4893 unique dishes, and therefore, the

resulting adjacency matrix is a square matrix with dimensions 4893x4893. Each element adjacency matrix [i , j] is 1 if there is a directed edge from node i to node j, and 0 otherwise.

3.2 HARDWARE REQUIREMENTS

The model was trained and tested on a dataset containing 4896 training data 612 testing data on Google Colab. The following hardware was used for the project:

1. **GPU** (Optional but highly recommended for deep learning):
   - A high-performance GPU, such as NVIDIA GeForce or Tesla series with 16 GB of GPU memory, significantly accelerates training times for deep neural networks.

2. **RAM:**
   - A minimum of 12 GB RAM is recommended, but for larger datasets and more complex models, 16 GB or more may be beneficial.

3. **STORAGE:**
   - Adequate storage space for the dataset, models, and other project-related files. SSDs are preferable for faster data access during training and inference.

4. **INTERNET CONNECTION:**
   - A reliable internet connection is necessary for downloading datasets, pre-trained models, and software libraries. It should be noted that the hardware specifications provided by Google Colab are subject to change based on the current availability of resources. Also, the runtime is limited

to a maximum of 12 hours, after which the runtime is disconnected. The user interface was implemented using the gradio package of python in Google Colab app, which allows running Jupyter notebooks as web apps and provides a simple way to share the interface with others. Overall, the hardware specification provided by Google Colab was sufficient for trainingand testing the sentiment analysis model on a dataset of this size and implementing a graphical user interface.

3.3 SOFTWARE REQUIREMENTS

- TensorFlow
- Keras
- NumPy
- Pandas
- Matplotlib
- Gradio

The proposed framework uses the TensorFlow, Keras , NumPy, Matplotlib, Python, Gradio , Pandas .

1)TensorFlow

TensorFlow is a free and open-source software library for dataflow and differentiable programming that can be used to solve a wide range of problems. It's a symbolic mathematics library that's also used in machine learning applications including neural networks.

2)Keras

Keras is an open-source neural network library written in Python. It is designed allowing developers to quickly prototype and build deep learning models. Keras provides a high-level interface for building neural networks, with pre-defined layers and models that can be easily assembled

to create complex architectures. It supports a range of network types, including convolutional networks, recurrent networks, and combinations of the two.

3) NumPy

NumPy is a library for the Python programming language, adding support for large, multi-dimensional arrays and matrices, along with a large collection of high-level mathematical functions to operate on these arrays.

4) Matplotlib

Matplotlib is a comprehensive library for creating static, interactive, and animated visualizations in Python. It is widely used for generating plots, charts, histograms, and other types of visualizations for data analysis and presentation purposes.

5) Gradio

Gradio is an open-source Python library that allows developers to quickly create customizable UI components for their machine learning models. With Gradio, developers can build interactive web-based applications that enable users to input data, visualize model outputs, and explore different configurations of the model.

6) pandas

Pandas is a popular open-source Python library used for data manipulation and analysis. It provides easy-to-use data structures and functions to work with structured data, primarily tabular data like CSV files or SQL tables .

4. EXPERIMETAL SETUP

4.1 DATASET

We created a dataset ourselves that includes 5100 rows and 4 columns. Each row represents a different dish, and the columns suggest three possible combinations for each dish. The columns are named "Combination1," "Combination2," and "Combination3," and they provide suggestions for complementary dishes to go with the main dish. Our dataset encompasses 4893 unique dishes carefully selected to offer a diverse culinary experience. We looked into the dataset and found out which types of dishes are most common and which combinations are suggested most often. With this dataset, our recommendation system can give personalized suggestions based on what dish you choose.

```
df
```

	item	combination1	combination2	combination3
0	appam	stew	kadalacurry	chutney
1	porota	chickenchilly	kadaichicken	veg kurma
2	chapati	chickenchilly	kadaichicken	veg kurma
3	puri	baaji	green peas masala	sambar
4	rotti	chickenchilly	kadaichicken	chicken curry
...	...	...	...	...
5095	Cranapple gulumal	Red currant gulumal	Quandong gulumal	Nashi pear gulumal
5096	Jambul gulumal	Salal berry gulumal	Red currant gulumal	Jaboticaba gulumal
5097	Jabong gulumal	Tamarind gulumal	Salal berry gulumal	Kabosu gulumal
5098	Kumquince gulumal	Ugni (Chilean guava) gulumal	Tamarind gulumal	Longan gulumal
5099	Pearberry gulumal	Voavanga gulumal	Quandong gulumal	Marang gulumal

5100 rows × 4 columns

Fig 4.1 Samples from dataset

During training, while downsampling process, it's possible to encounter situations where the dimensions become odd, potentially causing compatibility issues with subsequent layers, particularly with certain types of upsampling techniques. To address this, it's common to add extra columns or rows to the feature maps, ensuring that the dimensions remain even (Zero padding) . By doing so, the network can maintain consistency in feature map dimensions, facilitating smooth information flow and computation throughout the network.

4.2 LOSS FUCNTION AND OPTIMIZER USED

- MSE (mean squared error):

The use of MSE loss is common in regression tasks. In the context of autoencoders, which are used for unsupervised learning and reconstruction tasks, MSE loss measures the mean squared difference between the input data and the reconstructed output. It quantifies how well the autoencoder is able to reproduce its input.

Mathematically, for each data point, MSE is calculated as the average of the squared differences between the corresponding elements of the input and output vectors. The formula for MSE is :

MSE $= n1\sum I = 1n\ (yi - y\hat{}i)\ 2$

Where:

n is the number of data points.

yi is the actual value of the data point.

$y\hat{}i$ is the predicted (reconstructed) value of the data point.

In the context of autoencoders, minimizing the MSE loss encourages the model to learn a representation of the input data that captures its essential features, as the model is penalized for deviations between the input and the reconstructed output.

- Adam (Adaptive Moment Estimation):

This is the optimizer used in our model. It computes adaptive learning rates for each parameter based on the first and second moments of the gradients. Adam is a popular optimization algorithm commonly used in deep learning. It combines ideas from RMSprop and Momentum optimization methods.

4.3 DATASET TRAINING

We've passed our preprocessed training dataset into the model for training, which includes 4896 samples along with their labels. The training process is iterated over the same training dataset for 20 epochs (after it causing over-fitting) with the goal of improving accuracy of the model.

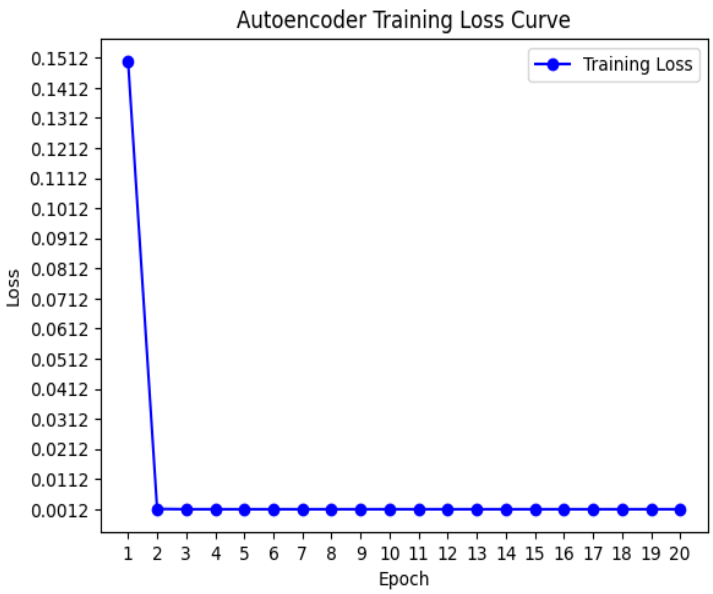


Fig 4.2 Diagram for training loss

5.RESULT ANALYSIS

Testing the model for accuracy is an important task. It's crucial for ensuring its accuracy and reliability in making predictions. It is also important in evaluating the performance, generalization, error analysis, selecting the best model and ensuring readiness for deployment

Performance evaluation: Testing allows one to evaluate the performance of the model and measure how accurate its predictions are. One can compare the predicted sentiment with the actual sentiment of a product review and see how often the model gets it right or wrong.

Generalization: Testing also helps one to determine how well the model will perform on new, unseen data. If the model is overfitting to the training data, it may not generalize well to new data. Testing allows one to detect overfitting and adjust the model accordingly.

Model selection: Testing multiple models can help one to choose the best one for the specific use case. One can compare the performance of different models and select the one that performs the best on the test data.

Error analysis: Testing can also help to identify the specific errors that the model is making and understand the reasons behind them. This can help the developer to fine-tune the model and improve its accuracy.

Deployment readiness: Testing is essential before deploying the model in a production environment to ensure it's robust, reliable and works as expected .

5.1 PRECISION

Precision is a metric that quantifies the accuracy of positive predictions made by the model. It is calculated as the ratio of true positive predictions to the total number of positive predictions made by the model. A high precision value indicates that the model makes fewer false positive predictions.

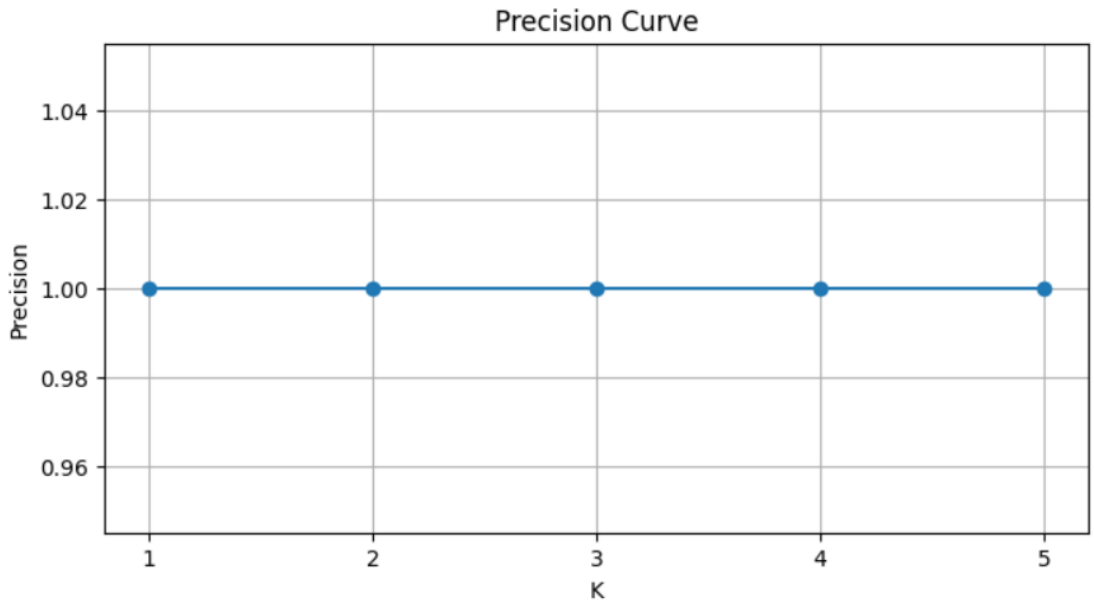


Fig 5.1 Precision curve

$$\text{Precision at k} = \frac{\text{Number of relevant items in k}}{\text{k}}$$

This equation calculates the proportion of relevant items among the top k items in the ranked list.

The 'k' refers to the number of items being considered in a ranked list.

5.2 RECALL

Recall, also known as sensitivity or true positive rate, measures the proportion of actual positive instances that are correctly predicted by the model. It is calculated as the ratio of true positive predictions to the total number of actual positive instances. A high recall value indicates that the model captures a larger proportion of positive instances.

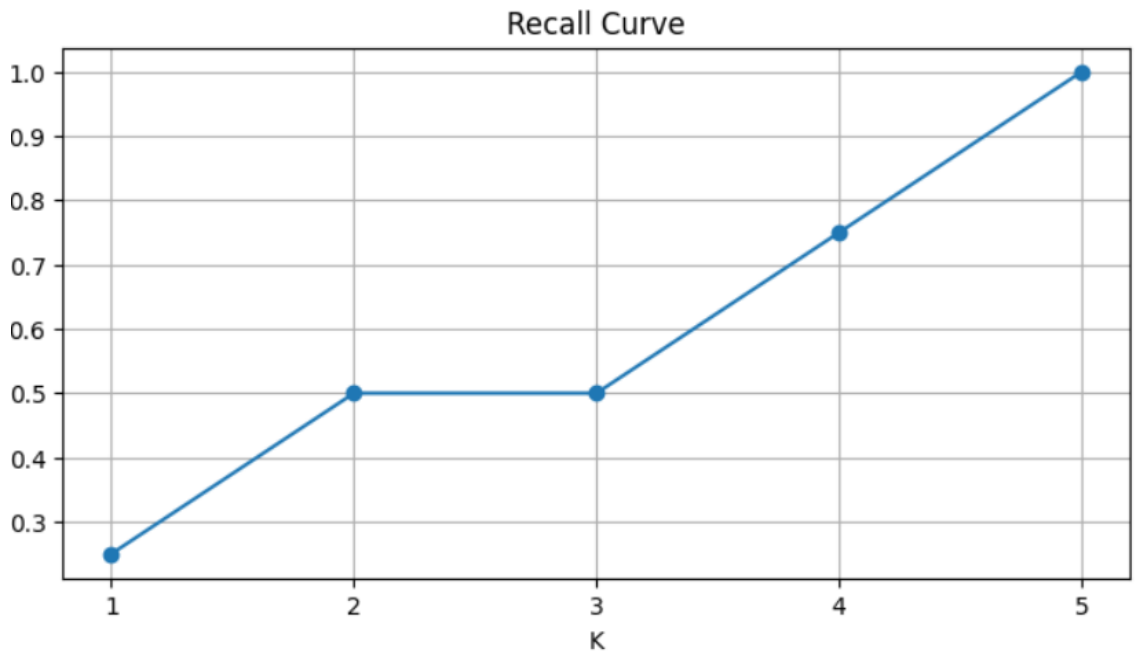


Fig 5.2 Recall curve

$$\text{Recall at k} = \frac{\text{Number of relevant items in top K}}{\text{Total number of relevant items}}$$

This equation calculates the proportion of relevant items that have been retrieved in the top k items of the ranked list.

6. IMPLEMENTATION

6.1 Gradio

Here Gradio is used for developing the graphical user interface. Here a GUI is developed using gradio which takes an image as input, then calls the predict function provided for prediction based on our model and gives the class name as output.

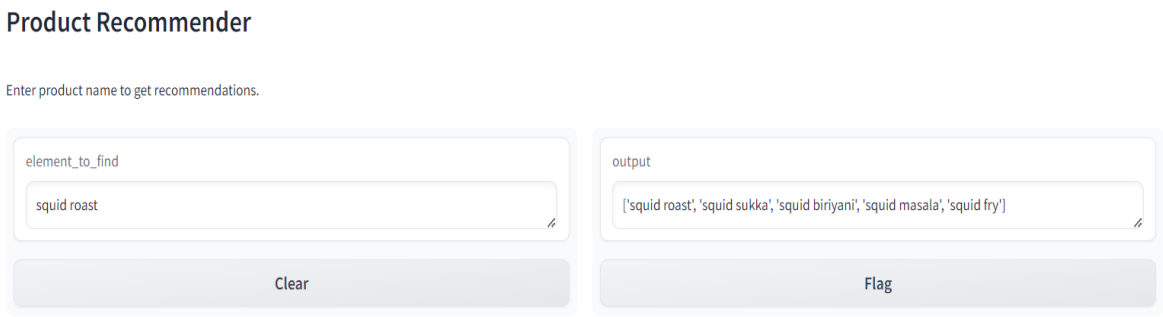


Fig 6.1 Graphical user interface

6.2 Webpage

Our smart restaurant webpage likely features a login page where user can authenticate themselves, accessing features such as personalized recommendations. The menu page would likely showcase restaurant's offerings with price. Our webpage aims to provide a convenient and enjoyable way for customers to explore restaurant's offerings.

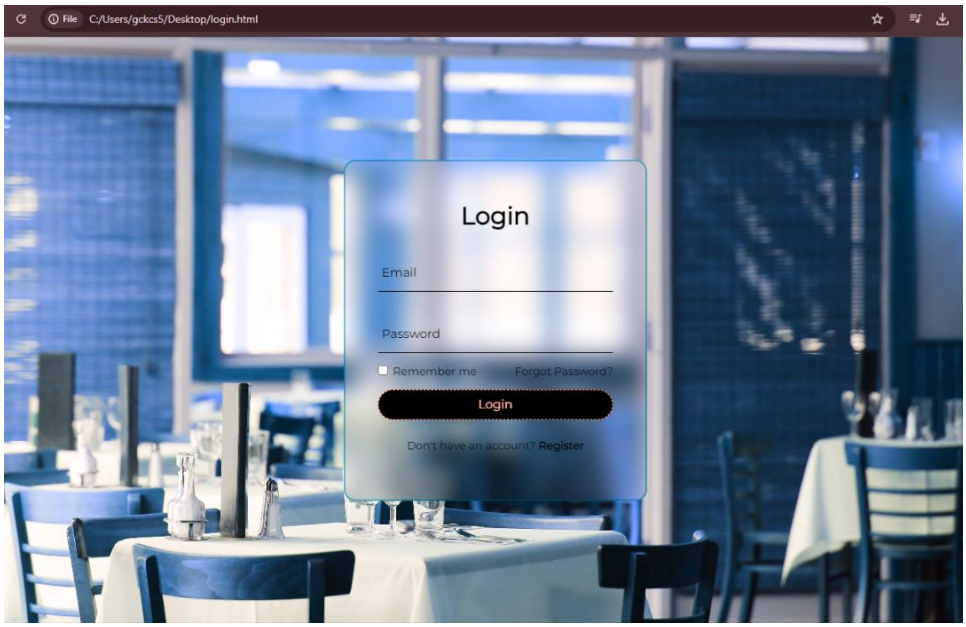


Fig 6.2 Login page

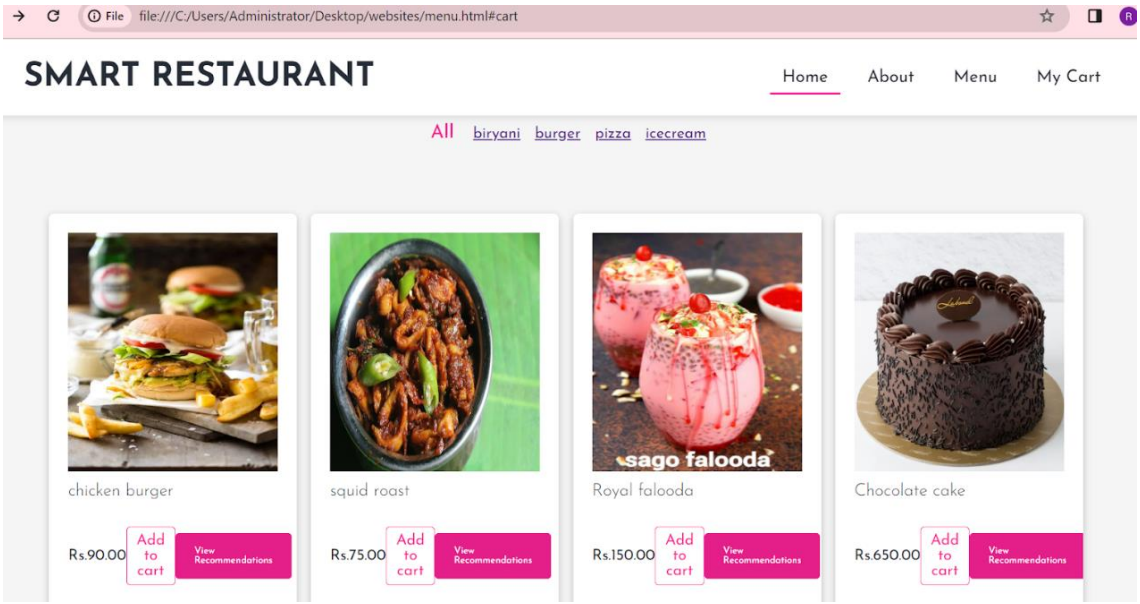


Fig 6.3 Restaurant menu

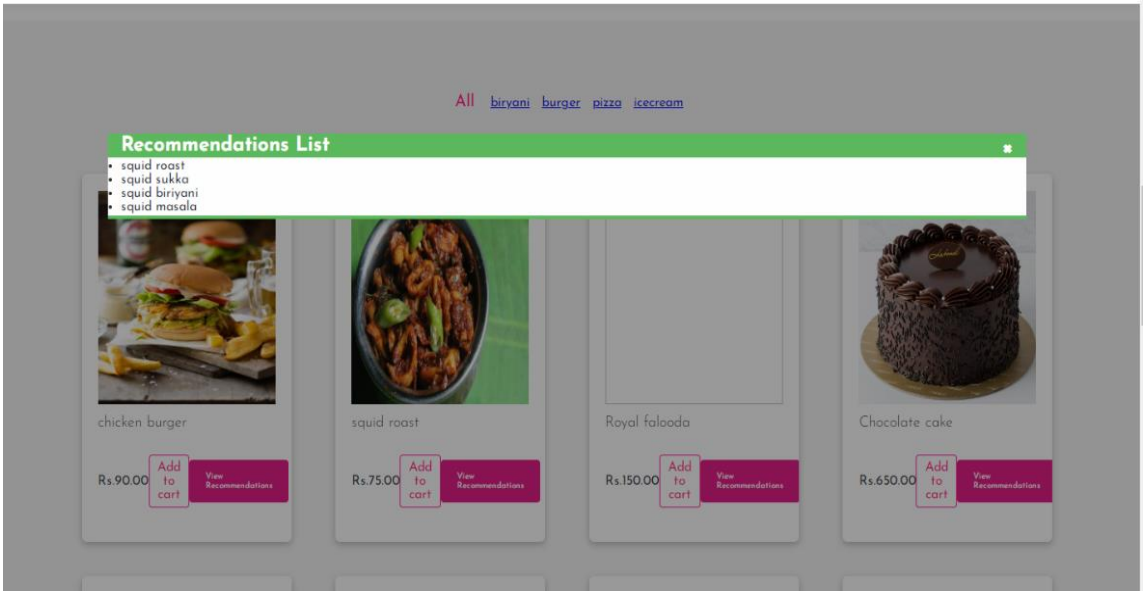
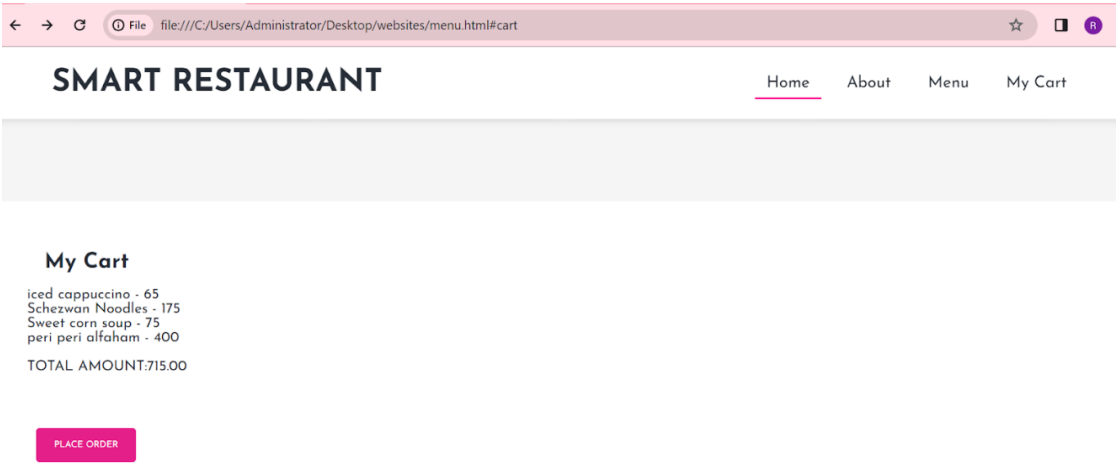


Fig 6.4 Recommendation page



THANK YOU

Fig 6.5 My cart

7.APPLICATION

- PERSONALISED RECOMMENDATIONS:

Utilizing machine learning algorithms to analyze customer preferences and behavior to provide personalized menu recommendations. This can enhance the customer experience and increase customer satisfaction.

- AUTOMATED ORDERING SYSTEM:

Implementing an AI-powered ordering system where customers can interact with a chatbot or voice assistant to place orders. Natural Language Processing (NLP) can be used to understand and process customer requests.

- TABLE MANAGEMENT:

Implementing AI for efficient table management, reservation handling, and waitlist optimization. This ensures that the restaurant operates smoothly during peak hours.

8.FUTURE WORKS

- **Enhanced personalization:**

Implement more advanced recommendation systems that take into account not only past order history but also real-time contextual information, user preferences, and dietary restrictions for highly personalized suggestions.

- **Feedback analysis with NLP:**

Use NLP techniques to analyze customer reviews and feedback more comprehensively. Understand sentiment, identify specific aspects of the restaurant experience that customers appreciate or dislike, and use this information for continuous improvement.

- **Augment Reality (AR) Menu Experience:**

Develop an AR-based menu experience for customers. Allow them to visualize menu items through AR, see nutritional information, and get additional details about the dishes.

9.CONCLUSION

Our Smart Restaurant project pioneers a new era in dining experiences, harnessing AI offer personalized recommendations. We redefine convenience and engagement in dining. We ensure a seamless end-to-end experience. Used convolutional autoencoder technology for recommendations. In today's fast-paced world, where personalized experiences are valued, our project stands as a beacon of innovation, showcasing the transformative power of AI in enhancing everyday interactions and shaping the future of dining.

10.REFERENCES

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